Formula Student Style Racecar Sensors

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Abstract*—This report reviews the sensors suited to such a car: IMU/INS, GPS/GNSS, wheel speed sensors, mono and stereo cameras, LiDAR and optical sensors. It then adds the supporting sensors that keep the car safe and controllable: temperature, pressure, position, current/voltage, load/strain, radar and ultrasonic. For each sensor it covers working principle, usecase for us, specifications, advantages and limitations, sensor fusion, and approximate cost and sourcing in India. It also covers the NVIDIA Jetson AGX Orin compute platform, steer-by-wire and brake-by-wire actuation, and the accelerometer and magnetometer types used in inertial units. A comparison table closes the report and recommends a core suite of LiDAR, camera, INS (or IMU plus RTK GNSS), Hall wheel speed sensors, actuator feedback sensors and the Jetson AGX Orin, with LiDAR giving cone position and the camera giving cone colour, and the INS, GNSS and wheel speed covering vehicle state. Temperature, pressure, steering-angle and current sensors form the essential safety and health layer around that core.

Keywords*—Formula Student, driverless, autonomous racecar, sensor fusion, IMU, INS, GNSS/RTK, wheel speed sensor, Hall effect, stereo camera, ZED 2i, LiDAR, Hesai Pandar40P, VectorNav VN-200, NVIDIA Jetson AGX Orin, steer-by-wire, brake-by-wire, temperature sensor, pressure sensor, position sensor, current sensor, load cell, radar, ultrasonic, SLAM, state estimation

I. Introduction

A Formula Student driverless (DV) car must complete Acceleration, Skidpad, Autocross and Trackdrive with no human input. The track is marked only by coloured cones, so the car has to do four jobs in a loop:

1) Perception: detect and classify cones (blue/yellow/orange) and estimate where they are.

2) Localisation and Mapping: know where the car is relative to the cones and build a map of the track.

3) State estimate: know velocity, yaw rate, slip and acceleration accurately, because planning and control depend on them.

4) Planning, Control and actuation: compute a path and speed profile, then command steering, throttle/torque and brakes, with an independent emergency brake system (EBS) as the safety net.

B. Sensor Fusion Architecture

1) Layers of fusion:

a) Time Synchronisation: a GNSS PPS pulse (or the VN-200 sync-out) time-stamps LiDAR, IMU and camera; use PTP/NTP on the Ethernet network.

b) Calibration: intrinsic calibration (camera), extrinsic calibration (LiDAR↔camera↔IMU↔vehicle frame), and lever-arm offsets for GNSS and IMU.

c) State estimation EKF/UKF: IMU as prediction; GNSS, wheel speed, steering angle and LiDAR odometry as measurements.

d) Perception fusion: LiDAR 3-D position + camera class/colour → coloured cone map.

e) SLAM: associate cones across laps (landmark-based graph SLAM) using the EKF pose as a prior.

2) Steer by wire/Brake by wire (acutaion): A driverless car needs electromechanical actuators that the autonomous system can command.

a) Steer-by-wire(streeing acutators):

- Working principle: a motor (BLDC or servo) with a gearbox drives the steering column or rack through a closed-loop position controller. A steering-angle sensor (absolute magnetic/Hall angle IC or optical encoder) feeds back angle.
- Feedback sensors: absolute angle encoder, motor current (Hall current sensor), end-stop/limit switches, redundant encoder for safety.
- Control: position loop on a real-time MCU; the planner's steering command goes over CAN.
- Issues: backlash, torque limits, response time (should track commands within tens of milliseconds), fail-safe behaviour, and the ability to disengage so the driver can steer when manual.

b) Brake-by-wire (service brake and EBS):

- Service brake actuator: an electric linear actuator or a servo-pressurised hydraulic system pushes the master cylinder in response to a brake-pressure target. A hydraulic pressure transducer gives closed-loop feedback.
- EBS (Emergency Brake System): by the rules, an independent, fail-safe system that applies the brakes if the autonomous system fails or shuts down. Teams commonly use a pneumatic actuator with a tank and a spring/pressure-based fail-safe, with pressure sensors monitoring the tank and brake line. Redundancy and monitoring (EBS must be checked at the start and often during the run) are rule-driven.
- Feedback sensors: brake line pressure (front/rear), tank pressure, actuator position, and sometimes pedal-position Hall sensors.

steering actuator ₹40,000–1.5 lakh (motor, gearbox, driver, encoder); brake actuator ₹30,000–1 lakh; pneumatic EBS components ₹40,000–1.2 lakh; pressure transducers ₹3,000–12,000 each; absolute magnetic encoders ₹500–5,000. Sources: Robu.in, Amazon.in, local pneumatic suppliers (Festo/SMC/Janatics distributors), Mouser/DigiKey India for sensors and ICs.

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II. Inertial Measurement Unit

An IMU/INS tells the car how it is moving (acceleration, rotation rate, orientation and, with GNSS aiding, velocity and position) at hundreds of hertz, which the controller and SLAM both depend on. This section follows the same six points as every other sensor, then covers the VectorNav VN-200 in detail.

A. Internal funtionlity and working principle

1) Gyroscope (Coriolis vibratory): a tiny proof mass is driven to vibrate. When the device rotates, the Coriolis force pushes the mass sideways; that sideways displacement is sensed (usually capacitively) and is proportional to angular rate.

2) Accelerometer (captive MEMS): a proof mass on springs moves under acceleration, changing the gap between capacitor plates. The capacitance change is converted to acceleration.

TABLE I. Types of Accelerometer

Types	***Principle***
Capacitive MEMS	Proof-mass motion changes capacitance
Piezoresistive	Strain in a beam changes resistance
Piezoelectric	Crystal generates charge under force
Thermal (convective)	Heated gas bubble shifts with acceleration
Servo / force-balance	Feedback coil holds the mass at null
Resonant / optical	Frequency or light shift with load

3) Magnetometer: measures the Earth's magnetic field to give an absolute heading reference.

TABLE II. Types of Magnetometer

Types	***Principle***
Hall effect	Voltage across a current-carrying plate in a magnetic field
AMR (anisotropic magnetoresistive)	Resistance of a permalloy strip changes with field direction
GMR / TMR	Spin-dependent resistance in thin-film stacks
Fluxgate	Saturating core driven by AC; field detected via harmonics
Lorentz-force MEMS	Lorentz force on a MEMS conductor
Search-coil, optically pumped, SQUID	Induction / atomic / superconducting

4) Processing: an on-board microcontroller applies factory calibration (temperature, misalignment, scale factor), time-stamps data and runs a fusion filter.

Gyro bias drifts, so integrating rate alone gives heading that wanders; that is why an INS aids the IMU with GNSS.

B. Use case and external Functionality

1) Yaw rate, lateral/longitudinal acceleration and attitude for state estimation, slip-angle estimation and the controller (pure pursuit, Stanley or MPC).

2) High-rate motion prediction between GNSS fixes and between LiDAR frames.

3) Motion compensation (deskewing) of each LiDAR scan so cones do not smear at speed

4) Time reference: a sync-out pulse can time-stamp LiDAR and camera frames.

5) Data logging for tuning and post-run analysis (g-g diagram, understeer, braking performance).

TABLE III. External Functionality

Aspect	***Detail***
Outputs	Raw gyro/accel/mag; orientation (quaternion or yaw-pitch-roll); body rates; NED velocity and position (INS); uncertainty/covariance estimates
Interfaces	UART (TTL or RS-232), SPI, sync in/out, event/trigger pins; ROS 2 drivers available
Configuration	Vendor software (e.g., VectorNav Control Center): output rate, filter tuning, mounting-frame rotation, GNSS antenna lever arm
Mounting	Rigid, close to the centre of gravity (or with a measured lever arm), axes aligned to the vehicle frame, away from motors, inverters and magnets, and on a stiff part of the chassis rather than a flexing panel
Power	3.3–5 V (check variant), low current, so it sits on the LV supply with a clean regulator

C. Specifications

A miniature GNSS-aided INS combining a MEMS IMU (3-axis gyro, accelerometer, magnetometer), a barometer, an onboard GNSS receiver and a built-in **EKF**. The IMU is sampled at high rate, the EKF propagates attitude, velocity and position, and GNSS measurements correct the drift. Fused data goes out over UART/SPI.

TABLE IV. Specifications in a VectorNav VN-200 (GNSS-aided INS)

Parameter	***Typical***
Sensors	3-axis gyro, 3-axis accel, 3-axis mag, barometer, GNSS
IMU output rate	up to 800 Hz
Filtered output rate	up to ~400 Hz
GNSS update	~5 Hz, single-frequency, multi-constellation
Gyro range / accel range	±2000 °/s / ±16 g
Pitch / roll accuracy	~0.03° RMS (dynamic)
Heading accuracy	~0.2° with good GNSS-aided dynamics (magnetic heading is worse)
Position accuracy	~2.5 m CEP (standard GNSS, no RTK)
Interfaces	UART (TTL/RS-232), SPI, sync in/out

D. Advntages nd Limitations

1) Advatages: very high update rate, works with no external signal, independent of lighting and weather, small and light, gives orientation (which LiDAR and camera do not give directly), and its raw data is the backbone of every vehicle-state estimator.

2) Limitations: drift (integration errors grow with time), bias that changes with temperature and warm-up, sensitivity to vibration and mounting, magnetometer disturbance from the car, and the need for calibration and correct frame alignment; without GNSS or another aiding source the position estimate degrades quickly.

E. Sensor Fusion

1) INS: IMU + GNSS: the EKF integrates IMU data at hundreds of Hz and corrects position, velocity and heading with GNSS at 5–20 Hz.

2) IMU + wheel speed + steering angle: a vehicle-model EKF improves longitudinal speed and slip estimation and bridges GNSS dropouts.

3) IMU + LiDAR: deskew each scan with the IMU, then use LiDAR-inertial odometry/SLAM (e.g., LIO-SAM style) so the IMU supplies orientation and short-term motion.

4) Time sync: use the GNSS PPS pulse or the unit's sync output; mis-timed IMU data corrupts the fusion more than noisy data does.

5) Calibration: measure the IMU-to-vehicle rotation and the lever arms to the GNSS antenna and LiDAR.

6) AHRS: IMU + magnetometer + a filter, giving orientation (roll, pitch, heading).

F. Cost and Sourcing

TABLE V. APPROXIMATE COST IN RUPEES

Item	*Cost*
Hobby IMU boards (BNO055, ICM-42688, MPU-series)	₹1,500–5,000
Industrial IMU/AHRS (VN-100, Xsens MTi, SBG Ellipse class)	₹1–2.5 lakh
GNSS-aided INS (VN-200 class)	₹2.3–4 lakh

hobby boards from Robu.in, ThinkRobotics, Amazon.in, Mouser India, DigiKey India; industrial units direct from VectorNav, Xsens/Movella or SBG Systems or their authorised distributors in India. Ask each vendor about university and Formula Student discounts. Prices are approximate and unquoted.

III. GPS / GNSS

A. Internal funtionality and working principle

GNSS (GPS, GLONASS, Galileo, BeiDou, and India's NavIC/IRNSS on supporting receivers) works by trilateration: each satellite broadcasts its position and a precise time stamp. The receiver measures signal travel time from at least four satellites (the fourth solves the receiver clock error) and computes position and velocity.

1) Standard GNSS: pseudorange based; ~1.5–3 m accuracy in open sky.

2) RTK (real time kinematics): uses carrier-phase measurements plus correction data from a nearby base station to reach ~1–2 cm horizontal accuracy.

3) PPK: same idea but post-processed from logged data (good for logging and offline map building).

4) Doppler velocity: GNSS also measures speed directly from Doppler shift, which is usually very accurate and not affected by wheel slip.

5) Dual antenna: two antennas give true heading independent of motion and magnetic noise.

B. Use case for Us and External Funtionality

In driverless events the track is unknown, and teams are generally not supposed to rely on a pre-surveyed track map. By Checking the current rules, we can know how GNSS may be used. Even when absolute position is limited, GNSS-derived velocity and heading are very valuable for state estimation.

C. Specifications

TABLE VI. SPECIFICATIONS TO LOOK FOR

Parameter	*Typical (u–blox ZED-F9P class)*
Constellations	GPS, GLONASS, Galileo, BeiDou (multi-band L1/L2)
RTK accuracy	~1cm +1 ppm horizontal
Updated rate	Up to ~20 Hz for RTK navigation
Convergence	Seconds with a good base/NTRIP stream
Interfaces	UART, USB, I²C, SPI
Antenna	Multiband active antenna on a ground plane

D. Advantages and limitaions

1) Advantages: no drift, absolute reference, cm-level with RTK, cheap compared with INS, not affected by lighting.

2) Limitations: needs open sky (multipath near buildings, tents, tall structures at competition sites), low update rate versus IMU, RTK needs a base station or correction service and radio/NTRIP link, antenna placement on a carbon monocoque/roll hoop needs a proper ground plane, and cold-start time.

E. Sensor Fusion

Fuse with the IMU in an **EKF**: the IMU propagates the state at hundreds of Hz and GNSS corrects drift at 5–20 Hz. Add wheel speed and steering for dropouts. Use GNSS PPS for time sync.

F. Cost and Sourcing

TABLE VII. APPROXIMATE COST IN RUPEES

Item	*Cost*
u-blox ZED-F9P RTK board (e.g., ArduSimple simpleRTK2B, SparkFun)	₹20,000–35,000
Multi-band antenna (×2 for rover + base)	₹5,000–12,000 each
Base-station setup (second F9P + antenna + radio/laptop)	₹30,000–50,000
Basic u-blox NEO-M8N / M9N (no RTK)	₹800–3,000

Robu.in, ThinkRobotics, Amazon.in, and ArduSimple/SparkFun resellers for the F9P modules; Mouser India, DigiKey India and element14 India for bare u-blox modules. RTK corrections: run your own base station on site, or use a correction service (availability and cost of CORS/NTRIP networks in India varies, so confirm).

IV. Wheel Speed Sensors

A. Internal Functionality and Working Principle

When current flows through a thin conductor or semiconductor plate and a magnetic field is applied perpendicular to it, a small voltage appears across the plate's sides: the Hall voltage, $V_H = (I \cdot B)/(n \cdot q \cdot t)$, where I is current, B is flux density, n is carrier density, q is electron charge and t is thickness. A Hall IC adds an amplifier, temperature compensation and often a Schmitt trigger on the same chip.

A toothed (or magnetised) ring rotates with the wheel hub or drive shaft. The sensor, mounted on the upright, generates one pulse per tooth. Wheel speed = (pulse frequency ÷ teeth) × tyre circumference (or rolling radius).

TABLE VIII. Comman technologies

Technology	***How it works***
Hall Effect (Active)	Described above
Variable reluctance (inductive)	Moving tooth changes flux in a coil
Optical encoder	Slotted disc interrupts light
Motor Encoder /resolver (EV)	Sensor on motor shaft

B. Use case for us and External Funtionality

1) Longitudinl velocity: average of non-driven wheels is the cleanest at low slip.

2) Wheel slip: for traction control and torque vectoring.

3) Odometry: for the EKF and SLAM during GNSS dropouts or LiDAR degeneracy (e.g., long straights).

4) Cross-checking sensors: for safty monnitorinng.

C. Specifications

TABLE IX. Specifications to look for

Parameter	***Typical***
Output	Digital pulse (open-collector or push-pull)
Resolution	20–60 teeth per revolution (more is better at low speed)
Air gap	~0.5–2 mm
Supply	5 V or 12 V
Temperature	−40 to +125 °C
Sensor update	limited by pulse rate; very fast at racing speeds

D. Advntages and Limitations

1) Advantages: cheap, robust, high rate, independent of environment, zero-speed capable (Hall).

2) Limitations: measures wheel rotation, not true ground speed, so slip and tyre radius changes cause error under hard acceleration, braking and cornering; quantisation at low speed with few teeth; mounting tolerance (air gap) is critical.

E. Sensor Fusion

Fuse with the IMU and GNSS in the EKF; treat the wheel speed measurement noise as larger when slip is high (use a slip-dependent covariance). Average the non-driven (or lower-slip) wheels. Compare four wheels for yaw-rate cross-checks.

F. Cost and Sourcing

TABLE X. Approximate cost in Rupees

Item	***Cost***
Basic Hall switch (A3144, etc.)	₹20–100
Gear-tooth Hall sensors (Allegro/Honeywell/Infineon class)	₹500–3,000 each
Automotive ABS-type sensor	₹800–4,000 each
Toothed ring (machined, custom)	₹1,000–5,000

Robu.in and Amazon.in for basic Hall ICs; Mouser India, DigiKey India and element14 India for automotive-grade parts; local machine shops for tone rings.

V. Mono Camaras and Stereo Camaras

A. Internal Functionality and Working Principle

A camera focuses light through a lens onto a CMOS image sensor. Each pixel converts photons to charge, which is read out, digitised and processed by an image signal processor (ISP).

1) Shutter type: global shutter exposes all pixels at once (no distortion at speed); rolling shutter exposes row by row, causing skew on fast motion.

2) Monocular Camara: one lens. Depth is not measured directly; it is inferred from known object size (cone dimensions), ground-plane assumptions or learned depth.

3) Stereo Camara: two lenses a known baseline apart. Depth comes from disparity: $Z = f \cdot B / d$ (f focal length in pixels, B baseline, d disparity). Depth error grows roughly with Z^2, so stereo is accurate close up and degrades with distance.

B. Use case for us and External Funtionality

cone detection and colour classification (a LiDAR cannot tell blue from yellow cones, a camera can), plus stereo depth for close-range cone position and as a redundant range source; its IMU gives a secondary inertial stream. Run YOLO-type detectors on the Jetson.

C. Specifications

The ZED 2i is a stereo camera with a built-in IMU, barometer and magnetometer and an IP66 enclosure. It is a stereo camera, but you can also use a single lens/image stream as a mono camera (or run the left image only) when you want only colour/classification.

TABLE XI. SPECIFICATIONS TO LOOK FOR

Parameter	*Typical*
Sensors	Two 4 MP rolling-shutter sensors
Baseline	120 mm
Resolution / frame rate	up to 2208×1242 at 15 fps; 1080p at 30 fps; 720p at 60 fps; WVGA up to 100 fps
Lens options	2.1 mm (wide, ~110° H) or 4 mm (narrower, ~72° H)
Depth range	~0.3–20 m (usable range for cones is much shorter, see below)
Built-in sensors	IMU, magnetometer, barometer, temperature
Interface	USB 3.0
Software	ZED SDK (needs NVIDIA GPU/CUDA; runs on Jetson), ROS 2 wrapper
Enclosure	IP66, aluminium

D. Advntages and Limitations

1) Advantages: easy integration, good SDK, built-in depth and IMU, rugged, colour information, low cost compared with LiDAR.

2) Limitations: : rolling shutter distorts at high speed and vibration; depth accuracy drops quickly beyond ~10–15 m (illustrative: at 1080p with ~1000 px focal length, a 0.25 px disparity error gives ~0.2 m depth error at 10 m and ~0.5 m at 15 m); sensitive to sun glare, shadows, low light, rain and lens dirt; USB 3 cable length and connector vibration need care; needs a GPU.

3) Better-suited alternatives: global-shutter industrial cameras (Basler ace, FLIR Blackfly S, Arducam global-shutter modules) for mono perception; Stereolabs ZED X (GMSL2, global shutter) if you move to a Jetson with GMSL; Intel RealSense (active stereo) for short range.

E. Sensor Fusion

project LiDAR cone clusters into the image (needs accurate camera–LiDAR extrinsic calibration), take colour/class from the camera and 3-D position from LiDAR. Match by IoU or nearest-neighbour association and time-sync with PPS/PTP.

F. Cost and Sourcing

ZED 2i ≈ **US$450–550** (≈ **₹45,000–65,000** landed). Industrial global-shutter mono cameras ≈ **₹25,000–80,000** plus lens (₹5,000–20,000). Sources: Stereolabs' official store and its authorised resellers (check the reseller list for India), Amazon.in, Robu.in and ThinkRobotics for Jetson-ecosystem camera kits; Basler/FLIR via their Indian distributors.

VI. LIDAR

A. Internal Functionality and Working Principle

A spinning mechanical LiDAR fires short pulses of infrared light (typically 905 nm) and measures the time of flight (ToF): distance = $c \cdot \Delta t/2$. A rotating assembly of laser–detector pairs (channels) stacked at different vertical angles sweeps 360°, producing a 3-D **point cloud** with x, y, z and **intensity** (reflectivity) per point. Cones are bright, retro-reflective objects, so intensity helps detect them.

B. Use case for us and External Funtionality

Primary cone detection and ranging (accurate 3-D positions to ±cm), main input for LiDAR-based SLAM and mapping, and obstacle detection. Mount low and at the front, e.g., on the front wing/nose mount or roll hoop, at a height where many beams hit the cones (cones are ~30 cm tall, so a lower mount helps).

C. Specifications

A 40-channel mechanical LiDAR widely used in autonomous vehicles and Formula Student Driverless teams.

TABLE XII. SPECIFICATIONS IN A HESAI PANDAR 40P

Parameter	*Typical*
Channels	40
Wavelength	905 nm, Class 1 (eye safe)
Range	up to ~200 m (10% reflectivity)
Range accuracy	±2 cm (typical)
Horizontal FOV	360°
Vertical FOV	about +15° to −25° (non-uniform spacing, denser near the horizon)
Horizontal resolution	~0.2° at 10 Hz, ~0.4° at 20 Hz
Frame rate	10 or 20 Hz
Points per second	~0.72 million (single return)
Interfaces	Ethernet (UDP), GPS/PPS sync input
Power / weight	~12 V, ~10–15 W / ~1.5 kg
Ingress protection	IP67

D. Advntages and Limitations

1) Advantages: very accurate range, long range, independent of ambient light (works at night and in glare), 360° coverage, intensity data, mature drivers and ROS 2 support, PPS time sync.

2) Limitations: no colour (needs a camera for blue/yellow), sparse vertical resolution (few beams hit a small cone far away), mechanical spinning part, high cost, ~1.5 kg and power draw, rain/fog degrade returns, motion distortion at speed (needs IMU deskew), and the Pandar40P is an older model, so availability is mostly used units, sponsorship or newer Hesai products (check Hesai's current lineup such as the Pandar and OT series).

E. Sensor Fusion

deskew each scan using IMU/INS; run ground-plane removal, clustering and cone classification; fuse with camera colour; feed LiDAR odometry/SLAM (LOAM/LIO-SAM/GraphSLAM style) with IMU and wheel odometry.

F. Cost and Sourcing

historically ≈ **US$8,000–10,000** new (≈ **₹7–10 lakh**); used or sponsored units can be much cheaper. Hesai often supports Formula Student teams, so ask for a sponsorship/discount via Hesai's sales team or an authorised distributor in India.

Cheaper alternatives to evaluate: Ouster OS1-32/64, Livox Mid-360, Robosense Helios/Bpearl, Velodyne VLP-16 (second-hand), Slamtec/RPLidar for bench testing only.

VII. Optical Sensors

in the broadest sense, "optical" means any sensor that works with light, and LiDAR (active laser time-of-flight) and mono/stereo cameras (passive imaging) do fit that definition. This report treats them in their own sections (5 and 6) because the task lists them separately, and uses this section for the other light-based sensors.

A. Internal Functionality and Working Principle

TABLE XIII. Types of Optical Sensors

Optical sensor	***Principle***
Optical ground-speed / slip-angle sensor (e.g., Corrsys-Datron/Kistler Correvit)	Non-contact; correlates road-texture patterns or uses laser-Doppler to measure true ground speed and slip angle
Optical flow sensor (PMW3901/PAA5100-type)	Tiny camera tracks surface texture frame-to-frame
Optical (rotary) encoder	Light through slits on a disc
Infrared thermometer / thermal array (e.g., MLX90614/MLX90640)	Detects IR radiation
Laser / ToF distance sensor (VL53L1X, LiDAR-Lite)	Time-of-flight

B. Specifications

TABLE XIV. Specifications to look for

Items	***Cost***
Correvit-class optical ground-speed sensor	₹10–25+ lakh (rental or university loan is more realistic)
PMW3901/PAA5100 optical flow board	₹1,500–4,000
MLX90614 IR temp sensor	₹800–2,500
MLX90640 thermal array	₹6,000–12,000
VL53L1X ToF sensor	₹500–1,500

C. Advntages and Limitations

1) Advantages: non-contact, no wheel-slip errors (ground-speed sensors), high resolution, immune to magnetic noise.

2) Limitations: industrial ground-speed sensors are very expensive and heavy; optical flow and ToF are affected by sunlight, surface texture and dust; mounting height and alignment are critical.

D. Sensor Fusion

Ground-speed sensors are mostly used as **reference sensors** during testing to validate the velocity/slip estimate from IMU+GNSS+wheel speed. They are not essential for competition.

E. Cost and Sourcing

Robu.in, ThinkRobotics, Amazon.in, Mouser India, DigiKey India; Corrsys-Datron/Kistler via their regional distributors on request.

VIII. Comparison and Recommendation

A. Recommended sensor suite

1) LiDAR (Hesai Pandar40P or equivalent): for accurate cone position and SLAM

2) Stereo camera (ZED 2i))optionally one global-shutter mono camera): for long-range cone colour, fused with LiDAR

3) INS (VN-200) for state estimation; if budget is limited, use a mid-range IMU (VN-100 or an ICM-42688/BNO-class module) with a ZED-F9P RTK GNSS and run your own EKF.

4) Four Hall-effect wheel speed sensors (and motor encoders if EV).

5) Steering angle encoder and brake pressure sensors: for closed-loop steer-by-wire and brake-by-wire, plus the EBS monitoring sensors.

6) Jetson AGX Orin: as the compute platform with a separate real-time safety controller.

B. Why this combintion

1) LiDAR + camera: cover each other's weakness: LiDAR gives precise positions and works in any light, cameras give the colour needed to know which side of the track a cone is on.

2) INS + GNSS + wheel speed: gives redundant velocity and heading estimates; wheel slip, GNSS dropouts and IMU drift each get covered by another source.

3) Steering/brake feedback sensors: make the actuators controllable and the safety case auditable.

4) Jetson AGX Orin: has the GPU headroom for deep-learning cone detection and SLAM at low latency.

TABLE XV. COMPARISION TABLE

Sensor	***Environment sensitivity***	***Range /Accuracy***	***Rate***	***Priority for DV***
Hesai Pandar40P LiDAR	Low (rain/fog moderate)	~200 m, ±2 cm	10–20 Hz	Essential
ZED 2i stereo camera	High (glare, light, rolling shutter)	0.3–20 m (best < 10 m)	up to 100 fps (lower res)	Essential
Mono global-shutter camera	High	Inferred depth	30–120 fps	Recommended
VectorNav VN-200 INS	Low (mag disturbed)	0.03° pitch/roll, 2.5 m GNSS	400–800 Hz	Essential **(or cheaper IMU + F9P)**
GNSS RTK (ZED-F9P)	Needs open sky	1–2 cm (RTK)	5–20 Hz	Strongly recommended
Wheel speed (Hall)	Low	Depends on teeth	kHz pulses	Essential
Steering / brake feedback (Hall/pressure)	Low	n/a	1 kHz+	Essential
Optical ground-speed (Correvit)	Medium	0.1% speed	100+ Hz	Optional (borrow/rent)
Optical flow / ToF / IR	High	Low	Varies	Optional

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